Number Theory: Congruences

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Introduction 1

- Pseudo-random Numbers
  - Example: find the sequence of pseudorandom numbers by linear congruential method
    $x_{n+1} = (7x_n + 4)\ \boldsymbol{mod}\ 9,\ x_0 = 3$
    - $\boldsymbol{mod}$ is an operator very similar as the remainder operator
    - E.g. 101 = 11 · 9 + 2, so 101 ***mod*** 11 = 2

Introduction 2

- $x_1 = 7x_0 + 4 \boldsymbol{mod} 9 = 7 \cdot 3 + 4 \boldsymbol{mod} 9 = 25 \boldsymbol{mod} 9 = 7$,
  $x_2 = 7x_1 + 4 \boldsymbol{mod} 9 = 7 \cdot 7 + 4 \boldsymbol{mod} 9 = 53 \boldsymbol{mod} 9 = 8$,
  $x_3 = 7x_2 + 4 \boldsymbol{mod} 9 = 7 \cdot 8 + 4 \boldsymbol{mod} 9 = 60 \boldsymbol{mod} 9 = 6$,
  $x_4 = 7x_3 + 4 \boldsymbol{mod} 9 = 7 \cdot 6 + 4 \boldsymbol{mod} 9 = 46 \boldsymbol{mod} 9 = 1$,
  $x_5 = 7x_4 + 4 \boldsymbol{mod} 9 = 7 \cdot 1 + 4 \boldsymbol{mod} 9 = 11 \boldsymbol{mod} 9 = 2$,
  $x_6 = 7x_5 + 4 \boldsymbol{mod} 9 = 7 \cdot 2 + 4 \boldsymbol{mod} 9 = 18 \boldsymbol{mod} 9 = 0$,
  $x_7 = 7x_6 + 4 \boldsymbol{mod} 9 = 7 \cdot 0 + 4 \boldsymbol{mod} 9 = 4 \boldsymbol{mod} 9 = 4$,
  $x_8 = 7x_7 + 4 \boldsymbol{mod} 9 = 7 \cdot 4 + 4 \boldsymbol{mod} 9 = 32 \boldsymbol{mod} 9 = 5$,
  $x_9 = 7x_8 + 4 \boldsymbol{mod} 9 = 7 \cdot 5 + 4 \boldsymbol{mod} 9 = 39 \boldsymbol{mod} 9 = 3$,
  $x_{10} = 7x_9 + 4 \boldsymbol{mod} 9 = 7 \cdot 3 + 4 \boldsymbol{mod} 9 = 25 \boldsymbol{mod} 9 = 7$,
  $x_{11} = 7x_{10} + 4 \boldsymbol{mod} 9 = 7 \cdot 7 + 4 \boldsymbol{mod} 9 = 53 \boldsymbol{mod} 9 = 8$,
  … …
- Single digit numbers will be generated in a fixed sequence
- If we know the formula, we know how they are generated

Introduction 3

- Congruence is very important in CS
  - Pseudo-random numbers, cryptography, hashing, etc.
- Topic plan
  - Divisibility and Modular Arithmetic
  - Primes and Greatest Common Divisors (gcd)
    - Pairwise Relatively Prime
  - How to solve congruences
- Topic flow

Divisibility { Congruence; Primes, GCD } Relatively Prime } Solve Congruence

Divisibility and Modular Arithmetic 1

- **Definition**: If *a* is an integer and *d* a positive integer, then there are unique integers *q* and *r*, with $\mathbf{0} \leq \boldsymbol{r} < \boldsymbol{d}$, such that $\boldsymbol{a} = \boldsymbol{dq} + \boldsymbol{r}$.
  - *d* is called the ***divisor***.
  - *a* is called the ***dividend***.
  - *q* is called the ***quotient***.
  - *r* is called the ***remainder***.

Definitions of Functions
div and **mod**
q = *a* **div** *d*
r = *a* **mod** *d*

- **Definition**: If a and b are integers with a ≠ 0, then a divides b if there exists an integer c such that b = ac.
  - The notation ***a* | *b*** denotes that *a* divides *b*.
  - 3 ∤ 7, 3 | 12

Divisibility and Modular Arithmetic 2

- **Example:**
  - What are the quotient and remainder when −11 is divided by 3?
  - **Solution**: when −11 is divided by 3, the quotient is −4 = −11 **div** 3, and the remainder is 1 = −11 **mod** 3.
    - Recall: $\mathbf{0} \leq \boldsymbol{r} < \boldsymbol{d}, \mathbf{0} \leq \boldsymbol{r} < \mathbf{3} \therefore \boldsymbol{r} \neq -\mathbf{2}$
- ***Note***: $a \boldsymbol{mod} m = a - m\lfloor a/m \rfloor$
- ***Remark***: % is used in C/C++/Java for modular arithmetic, and when $a < 0$, $a \% m \leq 0$

Divisibility and Modular Arithmetic 3

- Sometime we only care about the remainder
  - What is the time if it's 50 hours from now?
- Congruence
  - If *a* and *b* are integers and *m* is a positive integer, then *a* is ***congruent to b modulo m*** if *m* divides *a* − *b*.
  - The notation $\boldsymbol{a \equiv b \pmod{m}}$ says that *a* is congruent to *b* modulo *m* (or *a* and *b* are congruent modulo *m*).
  - We say that $\boldsymbol{a \equiv b \pmod{m}}$ is a ***congruence*** and that *m* is its ***modulus***.

Divisibility and Modular Arithmetic 4

- **Example**:
  - Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.
- **Solution**:
  - 17 ≡ 5 (mod 6) because 6 divides 17 − 5 = 12.
  - 24 ≢ 14 (mod 6) since 24 − 14 = 10 is not divisible by 6.
- **Discoveries**
  - $\boldsymbol{a \equiv b\ (mod\ m)}$ iff $a\ \boldsymbol{mod}\ m = b\ \boldsymbol{mod}\ m$ Easy to understand
    - 17 mod 6 = 5, 11 mod 6 = 5
  - $\boldsymbol{a \equiv b\ (mod\ m)}$ iff $\exists k \in \mathbb{Z}\ (a = b + km)$ Easy for proof
    - 17 ≡ 11 (mod 6) because 6 divides 17 − 11 = 6.
    - 17 = 2 · 6 + 5, 11 = 1 · 6 + 5, and 17 = 11 + 1 · 6
- ***Note***: $\boldsymbol{a \equiv b\ (mod\ m)} \leftrightarrow \boldsymbol{\exists k \epsilon \mathbb{Z}(a = b + km)} \leftrightarrow$ $a\ \boldsymbol{mod}\ m = b\ \boldsymbol{mod}\ m$

Divisibility and Modular Arithmetic 5

- Do more modular arithmetic
  - If $a \equiv b(mod\ m)$ and $c \equiv d(mod\ m)$, then $a + c \equiv b + d(mod\ m)$ and $ac \equiv bd(mod\ m)$
    - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$
    - $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$
    - Recall: $\boldsymbol{a \equiv b\ (mod\ m)}$ iff $\exists k \in \mathbb{Z}\ (a = b + km)$
  - **Example**: $7 \equiv 2(mod\ 5)$ and $11 \equiv 1(mod\ 5)$
    - $7 + 11 \equiv 2 + 1(mod\ 5)$ → $18 \equiv \mathbf{3}\ (\boldsymbol{mod}\ \mathbf{5})$
    - $7 \cdot 11 \equiv 2 \cdot 1\ (mod\ 5)$ → $77 \equiv \mathbf{2}\ (\boldsymbol{mod}\ \mathbf{5})$
  - **Example**: $7 \equiv 2(mod\ 5)$, $8 \equiv 8(mod\ 5)$, $3 \equiv 3(mod\ 5)$
    - $7 - \mathbf{8} \equiv 2 - \mathbf{8}\ (mod\ 5)$ → $-1 \equiv \mathbf{-6}\ (\boldsymbol{mod}\ \mathbf{5})$
    - $7 \cdot \mathbf{3} \equiv 2 \cdot \mathbf{3}\ (mod\ 5)$ → $21 \equiv \mathbf{6}\ (\boldsymbol{mod}\ \mathbf{5})$

Primes + Greatest Common Divisors 1

- **Definition**: A positive integer *p* greater than 1 is called *prime* if the only positive factors of *p* are 1 and *p*. A positive integer that is greater than 1 and is not prime is called *composite*.
  - **Example**:
    - The integer 7 is prime because its only positive factors are 1 and 7, but 9 is composite because it is divisible by 3.

Primes + Greatest Common Divisors 2

- **Definition**: prime factorization
  - Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.
  - **Examples**:
    - $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
    - $641 = 641$
    - $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
    - $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

Primes + Greatest Common Divisors 3

- **Definition**: Let *a* and *b* be integers, not both zero. The largest integer *d* such that $d \mid a$ and also $d \mid b$ is called the greatest common divisor of *a* and *b*. The greatest common divisor of *a* and *b* is denoted by gcd(a,b).
  - $\gcd(24, 36) = 12$
  - $\gcd(17,22) = 1$
- Find gcd by prime factorization
  - **Example**: find $\gcd(120, 500)$
  - **Solution**: $120 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 5 = 2^3 \cdot 3 \cdot 5$

    $500 = 2 \cdot 2 \cdot 5 \cdot 5 \cdot 5 = 2^2 \cdot 5^3$

    common prime factors and exponents: 2^2 and 5

    $\gcd(120, 500) = 2^2 \cdot 5 = 20$

Primes + Greatest Common Divisors 4

- **Definition**: The integers $a_1, a_2, \ldots, an$ are *pairwise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.
  - **Example**: Because $\gcd(7,3) = 1$, $\gcd(3,26) = 1$, and $\gcd(26,7) = 1$, then 7, 3, and 26 are pairwise relatively prime.
- Pairwise relative prime is commonly used in linear congruential function
  - Recall the introduction: Pseudo-random numbers

Primes + Greatest Common Divisors 5

- **Euclidean Algorithm**
  - Let $a = bq + r$, where *a*, *b*, *q*, and *r* are integers. Then $\gcd(a,b) = \gcd(b,r)$.
    - If $c|a$ and $c|b$, then $c|bq$, and thus $c|r$ since $a - \mathrm{bq} = \mathrm{r}$
    - If $c|r$ and $c|b$, then $c|bq$, and thus $c|a$ since $r + bq = a - \mathrm{bq} + \mathrm{bq} = \mathrm{a}$

```
procedure gcd(a, b: positive integers)
x := a
y := b
while   y ≠ 0
        r := x mod y
        x := y
        y := r
return x      {gcd(a,b) is x}
```

Replace the big number by a smaller number in gcd.

Primes + Greatest Common Divisors 6

- **Example**: Find gcd(91, 287):
- $287 = 91 \cdot 3 + 14$ Divide 287 by 91, remainder is 14
- $91 = 14 \cdot 6 + 7$ Divide 91 by 14, remainder is 7
- $14 = 7 \cdot 2 + 0$ Divide 14 by 7, remainder is 0

Stopping condition

gcd(287, 91) = gcd(91, 14) = gcd(14, 7) = 7

- All the equations above are all linear equations
- Do you think gcd(287,91) is a linear combination of 287 and 91 ?
  - gcd(287,91) = 287p + 91q (p and q are integers)
  - Try to reverse the procedure:
    - $14 = 287 - 91 \cdot 3$
    - $7 = 91 - 14 \cdot 6 = 91 - (287 - 91 \cdot 3) \cdot 6 = -6 \cdot 287 + 19 \cdot 91$

Solve Congruences 1

- The basic way to encrypt a text message is to use a linear congruential function

$$f_p = (a \cdot p + b)\ mod\ 26$$

where a and b are integers, and $\gcd(26, a) = 1$ (26 and a are relatively prime).

- **Example**: we can use $f_p = (7p + 3)\ mod\ 26$ to encrypt letter K which can be represented as 10, and $f(10) = (7 \cdot 10 + 3)\ mod\ 26 = 21$ which represents V, so K is encrypted as V!
  - This is a cool technique: try to encrypt “I miss u” :-D

Solve Congruences 2

- To decrypt the message we need to solve the linear congruence used in the encryption

- To make the linear congruence solvable, typical we can choose pairwise relatively prime integers
  - Pairwise relatively prime makes this function one-to-one correspondence (invertible) !
    - Look at the pseudo-random numbers

Solve Congruences 3

- Recall
  - K is 10, so $p = 10$
  - $f(p) = (7p + 3)\ mod\ 26 = (7 \cdot 10 + 3)\ mod\ 26 = 21$
  - 21 is V
- If we see the encrypted message, e.g. V, how can we get the original message back?
  - How to solve linear congruence
    $(7p + 3) \equiv 21\ (mod\ 26)$
- Can we divide both side by 7?
  - No! we can't! Sadly, there is no such rule to do this.
  - But we can multiply a number to each side! (slide 9)

Solve Congruences 4

- To solve $(7p + 3) \equiv 21\ (mod\ 26)$
  - The solution should look like $p \equiv a\ (mod\ 26)$, where a is an integer in [0,25]
  - We can multiply both side by 15

Steps	Results	Explanations
1	$(7p + 3) \equiv 21\ (mod\ 26)$	Original congruence
2	$7p \equiv 18\ (mod\ 26)$	-3 for both sides
3	$15 \cdot 7p \equiv 15 \cdot 18\ (mod\ 26)$	*15 for both sides
4	$15 \cdot 7p \equiv 105p \equiv p + 26 \cdot 4p \equiv p\ (mod\ 26)$	Reorganize LHS in line 3
5	$15 \cdot 18 \equiv 270 \equiv 270 - 26 \cdot 10 \equiv 10(mod\ 26)$	Reorganize RHS in line 3
6	$p \equiv 10\ (mod\ 26)$	Link line 4 and 5, based on line 3

10 is K ! We get 10 back ! But why multiply 15?

Solve Congruences 5

- How can we get such a multiplier?
  - $(7p + 3) \equiv 21 \ (mod\ 26)$ → $7p \equiv 18 \ (mod\ 26)$
    - 7, 3, and 26 are pairwise relatively prime
    - gcd(26, 7) = 1
  - Look at the procedure in Euclidean Algorithm:
    - 26 = 7 · 3 + 5 ①
    - 7 = 5 · 1 + 2 ②
    - 5 = 2 · 2 + 1 ③

$$\mathbf{t} \cdot 7p \equiv 1 \cdot p \equiv \boxed{p \equiv \mathbf{t} \cdot 18 \ (mod\ 26)}$$

$$\mathbf{t} \cdot 7 \equiv 1 \ (mod\ 26)$$

$$\gcd(26,7) = 1 = s \cdot 26 + \mathbf{t} \cdot 7$$

  - Trace back: slide 15 ---- gcd is a linear combination of 7 and 26

Steps	Results	Explanations
1	$1 = 5 - 2 \cdot 2$	Reorganize ③
2	$2 = 7 - 5 \cdot 1$	Reorganize ②
3	$1 = 5 - 2 \cdot 2 = 5 - (7 - 5) \cdot 2 = -2 \cdot 7 + 3 \cdot 5$	Line 2 into line 1
4	$5 = 26 - 7 \cdot 3$	Reorganize ①
5	$1 = -2 \cdot 7 + 3 \cdot 5 = -2 \cdot 7 + 3 \cdot (26 - 7 \cdot 3) = 3 \cdot 26 - 11 \cdot 7$	Line 4 into line 3

Solve Congruences 6

- How can we get such a multiplier? (cont'd)
  - So we know we can multiply -11 to both sides of $(7p + 3) \equiv 21 \ (mod\ 26)$ to get $p \equiv 10 \ (mod\ 26)$:

$(3 \cdot 26 - 11 \cdot 7) \bmod 26 = (-11 \cdot 7) \bmod 26 = 1 = \gcd(26,7)$

  - And $-11 \equiv 15 \ (mod\ 26)$, so we can also multiply 15 (look at slide 9): $-11 \cdot 7p \equiv (-11 + 26) \cdot 7p \equiv 15 \cdot 7p$
  - The secret of such a multiplier
    - 15 is called an inverse of 7 modulo 26, since

$$15 \cdot 7 \equiv 1 \ (mod\ 26)$$

      - Similar as 1/3 is the inverse of 3, since 1/3 * 3 = 1
    - $15 \cdot 7 \equiv 1 \ (mod\ 26)$, then $15 \cdot 7p \equiv p \ (mod\ 26)$
    - By multiplication, we can simplify the LHS to p, and thus the result would be $p \equiv 10 \ (mod\ 26)$

Practice 1

- 1. Evaluate:
  - 155 mod 19
  - -97 mod 11

Practice 2

- 1. Evaluate:
  - 155 mod 19
    - 155 = 19 * 8 + 3, so 155 mod 19 = 3
  - -97 mod 11
    - -97 = 11 * (-9) + 2, so -97 mod 11 = 2.

Practice 3

- 2. Suppose a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integer c with $0 \leq c \leq 12$ such that
  - $c \equiv 9a \pmod{13}$
  - $C \equiv 2a + 3b \pmod{13}$

Practice 4

- 2. Suppose a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integer c with $0 \le c \le 12$ such that
  - $c \equiv 9a \pmod{13}$
    - $c \equiv 9a \equiv 9*4 \equiv 36 \equiv 13 * 2 + 10 \equiv 10 \pmod{13}$
  - $c \equiv 2a + 3b \pmod{13}$
    - $c \equiv 2a + 3b \equiv 2*4 + 3*9 \equiv 35 \equiv 13 * 2 + 9 \equiv 9 \pmod{13}$

Practice 5

- 3. Find the greatest common divisors of the pairs of integers
  - $3^7 \cdot 5^3 \cdot 7^3$, and $2^{11} \cdot 3^5 \cdot 5^9$
  - $3^{13} \cdot 5^{17}$, and $2^{12} \cdot 7^{21}$

Practice 6

- 3. Find the greatest common divisors of the pairs of integers
  - The key is to find the shared prime divisors
  - $3^7 \cdot 5^3 \cdot 7^3$, and $2^{11} \cdot 3^5 \cdot 5^9$
    - $3^5 \cdot 5^3$
  - $3^{13} \cdot 5^{17}$, and $2^{12} \cdot 7^{21}$
    - 1

Practice 7

- 4. Use the Euclidean algorithm to find
  - gcd(1529, 14038)

Practice 8

- 4. Use the Euclidean algorithm to find
  - gcd(1529, 14038)
    =gcd(1529, 14038 mod 1529)
    =gcd(1529, 277)
    =gcd(277, 1529 mod 277)
    =gcd(277, 144)
    =gcd(144, 277 mod 144)
    =gcd(144, 133)
    =gcd(133, 144 mod 133)
    =gcd(133, 11)
    =gcd(11, 133 mod 11)
    =gcd(11, 1)
    =1

How about gcd(1529, 14039)? 139

Practice 9

- 5. Find an inverse of a modulo m for the pair of relatively prime integers
  - a = 55, m = 89

Practice 10

- 5. Find an inverse of a modulo m for the pair of relatively prime integers
  - a = 55, m = 89
    - gcd(55, 89)

=gcd(55, 34)	--- 89 = 55 * 1 + 34:	34 = 89 - 55
=gcd(34, 21)	--- 55 = 34 * 1 + 21:	21 = 55 - 34
=gcd(21, 13)	--- 34 = 21 * 1 + 13:	13 = 34 - 21
=gcd(13, 8)	--- 21 = 13 * 1 + 8:	8 = 21 - 13
=gcd(8, 5)	--- 13 = 8 * 1 + 5:	5 = 13 - 8
=gcd (5, 3)	--- 8 = 5 * 1 + 3:	3 = 8 – 5
=gcd(3, 2)	--- 5 = 3 * 1 + 2:	2 = 5 – 3
=gcd(2,1)	--- 3 = 2 * 1 + 1:	1 = 3 – 2

=1
= 3 - 2 = 3- (5-3)
= 5 * (-1) + 3 * 2 = 5 * (-1) + (8-5) * 2
= 8 * 2 – 5 * 3 = 8 * 2 – (13-8) * 3
= 13 * (-3) + 8 * 5 = 13 * (-3) + (21 - 13) * 5
= 21 * 5 – 13 * 8 = 21 * 5 – (34 - 21) * 8
= 34 * (-8) + 21 * 13 = 34 * (-8) + (55 - 34) * 13
= 55 * 13 – 34 * 21 = 55 * 13 – (89 - 55) * 21
= 89 * (-21) + 55 * 34
= 55 * s + 89 * t this is the linear combination we want: s = 34

Extension 1

- 1. Show that if a and b are congruent modulo 7, then 10a+13 and -4b+20 are also congruent modulo 7.
  - Look at slide 9.
- 2. $\gcd(2n+1, 3n+2) = ?$
  - Look at slide 14.
  - Similar as the steps in Euclidean Algorithm
  - 1

Extension 2

- 3.Hashing functions (page 304 in the textbook)
  - Page 287 in 7-th edition